

Expanded Control Layout Documentation

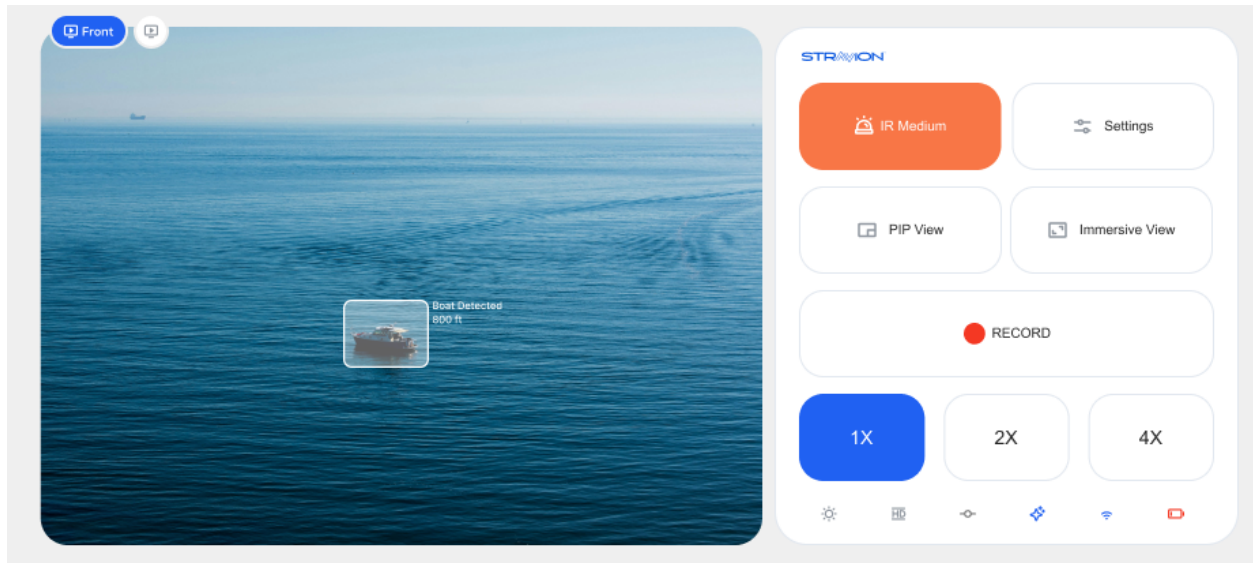


Figure 1: Expanded Layout

Overview

The Expanded Control Layout (`ExpandedControlLayout.kt`) provides a comprehensive control interface with full access to camera controls, recording functionality, and system status indicators. This layout is the most feature-rich interface, typically displayed when users need detailed control over the camera system.

Layout Structure

Header Section

- **Stravion Logo & Text:** Displays the application branding
 - Logo adapts to system theme (white in dark theme, gray in light theme)
 - Text size adjusts based on device type (20sp for tablet, 16sp for phone)

Control Button Rows

Settings Button

- **ID:** Settings
- **Function:** Opens the full settings/configuration panel
- **Visual Styling:**
 - **Dark Theme:**
 - * Background: Medium dark background (`AppColors.ButtonBgColor` - #333333)
 - * Icon Color: Medium gray (`AppColors.ButtonIconColor` - #808080)
 - * Border: Immersive button border (`AppColors.immersiveButtonBorderColor` - #E2E8F0)
 - **Light Theme:**
 - * Background: White (`AppColors.ButtonBgColor`)
 - * Icon Color: Medium gray (`AppColors.ButtonIconColor` - #AEAEAE)
 - * Border: Light gray border (#E2E8F0)
- **Click Action:** Triggers `onSettingsClick()` callback
- **Weight:** 1x width (standard button ratio)
- **Size:** 56dp (phone) / 112dp (tablet) when regular

- **Always Enabled:** Yes
- **API Used:** No direct API - UI navigation only

Picture-in-Picture Button

- **ID:** `picture-in-picture`
- **Function:** Reserved for picture-in-picture camera view functionality
- **Current Status:** **Disabled** - Not currently implemented
- **Visual Styling:**
 - **All Themes** (Disabled State):
 - * Background: Immersive button border color (`AppColors.immersiveButtonBorderColor` - `#E2E8F0`)
 - * Icon Color: Light gray (`#C5CBD4`)
 - * Border: Immersive button border (`#E2E8F0`)
 - * Enabled: False
- **Icon:** Picture-in-picture icon (`R.drawable.picture_in_picture_line`)
- **Click Action:** Currently empty placeholder (`{}`)
- **Weight:** 1x width (standard button ratio)
- **Size:** 56dp (phone) / 112dp (tablet) when regular
- **Layout Presence:** Available in expanded control layout button configurations
- **Future Implementation:**
 - Intended to provide split-screen or overlay camera view
 - May enable secondary camera feed or resizable camera window
 - Could integrate with Android's native Picture-in-Picture mode for camera feeds
 - Potential for multi-camera view management
- **API Used:** No current API - reserved for future implementation

IR (Infrared) Button

- **ID:** `ir`
- **Function:** Should cycle through four IR intensity levels for night vision
- **Current Implementation Issue:** Currently only toggles between off (0) and high (15), but needs to be enhanced
- **Required Enhancement:** Implement 4-level cycling with the following states:

Required Color States and Logic:

1. **Off** (brightness = 0):
 - Background: White (`Color.White`)
 - Icon: Medium gray (`AppColors.ButtonIconColor` - `#808080`)
 - Border: Immersive button border (`#E2E8F0`)
2. **Low** (brightness = 5):
 - Background: Light orange (`#FFA07D`)
 - Icon: White (`Color.White`)
 - Border: Light orange (`#FFA07D`)
3. **Medium** (brightness = 10):
 - Background: Medium orange (`#F87646`)
 - Icon: White (`Color.White`)
 - Border: Medium orange (`#F87646`)
4. **High** (brightness = 15):
 - Background: Dark orange (`#F55114`)
 - Icon: White (`Color.White`)
 - Border: Dark orange (`#F55114`)

Required Cycling Logic:

- **Click Sequence:** Off → Low → Medium → High → Off (repeats)
- **Current State:** `currentState.irBrightness` determines current level
- **State Mapping:**
 - `irBrightness = 0` → Off
 - `irBrightness = 5` → Low
 - `irBrightness = 10` → Medium
 - `irBrightness = 15` → High

Current Implementation (Needs Change):

- **Current Logic:** Simple toggle between 0 (off) and 15 (on)
- **Current Colors:** Only uses standard button colors and Record Red
- **Location:** `CameraControlViewModel.toggleIR()` method
- **Issue:** Does not support the 4-level cycling requirement

Disabled State (Vision Mode active):

- **Background:** Immersive button border color (`AppColors.immersiveButtonBorderColor` - #E2E8F0)
- **Icon:** Light gray (#C5CBD4)
- **Border:** Standard
- **Enabled:** False
- **Click Action:** `cameraControlViewModel.toggleIR()` (needs enhancement)
- **Enabled When:** `currentVisionMode != VisionMode.VISION`
- **Layout:** Row layout on tablets, Column on phones
- **API Used:**
 - **Primary API:** `MotocamAPIAndroidHelper.setIrBrightnessAsync()`
 - **Underlying Protocol:** `MotocamAPIHelper.setImgIRBrightnessCmd(value)`
 - **Command Structure:**

Request Packet:

Header	Command	SubCommand	Data Length	Data	CRC
SET(1)	IMAGE(4)	IRBRIGHTNESS(4)	1	0/5/10/15	1 byte

 - * Header: `Header.SET` (1)
 - * Command: `Commands.IMAGE` (4)
 - * SubCommand: `ImageSubCommands.IRBRIGHTNESS` (4)
 - * Values: 0 (off), 5 (low), 10 (medium), 15 (high)
 - **Success Response:**

Response Packet:

Header	Command	SubCommand	Data Length	Success Flag	Data	CRC
ACK(3)	IMAGE(4)	IRBRIGHTNESS(4)	2	Success(0)	Success(0)	1 byte
 - **Error Response:**

Response Packet:

Header	Command	SubCommand	Data Length	Failed Flag	Error Code	CRC
ACK(3)	IMAGE(4)	IRBRIGHTNESS(4)	2	Failed(1)	-1 to -6	1 byte
 - **HTTP Communication:** HTTP POST to `http://192.168.2.1:80/api/motocam_api`
 - **Content-Type:** `application/octet-stream`

- **Brightness Range:** 0-15 (discrete levels: 0, 5, 10, 15)

Collapse Screen Button

- **ID:** collapse-screen
- **Function:** Collapses the interface to minimal control layout
- **Visual Styling:**
 - **Dark Theme:**
 - * Background: Medium dark background (`AppColors.ButtonBgColor` - #333333)
 - * Icon Color: Medium gray (`AppColors.ButtonIconColor` - #808080)
 - * Border: Immersive button border (#E2E8F0)
 - **Light Theme:**
 - * Background: White
 - * Icon Color: Medium gray (#AEAFAE)
 - * Border: Light gray (#E2E8F0)
- **Click Action:** Triggers `onCollapseClick()` callback
- **Always Enabled:** Yes
- **API Used:** No direct API - UI state management only

Recording Control Section

Record Button

- **Function:** Starts/stops screen recording with automatic layout switching
- **Visual States:**
 - **Not Recording:**
 - * Background: Medium dark background (`AppColors.MediumDarkBackground` - #333333)
 - * Dot: Red variant color (`AppColors.RecordRed` - #F43823)
 - * Text: “RECORD” (medium light gray color - `AppColors.MediumLightGray`)
 - **Recording:**
 - * Background: Record red (`AppColors.RecordRed` - #F43823)
 - * Dot: White with pulsing animation (scale 1.0 to 1.2)
 - * Text: “RECORDING [duration]” (white color - #FFFFFF)
 - **Stopping Recording:**
 - * Background: Record red (70% opacity - `RecordRed.copy(alpha = 0.7f)`)
 - * Dot: White (70% opacity)
 - * Text: “STOPPING RECORDING...” (white color)
 - **Saved to Gallery:**
 - * Background: Green (#4CAF50)
 - * Dot: Green (#4CAF50)
 - * Text: “SAVED TO GALLERY” (white color)
- **Click Behavior:**
 - **When Not Recording:** First collapses to minimal layout, then starts recording
 - **When Recording:** Stops recording directly
- **Height:** 112dp on tablets, 56dp on phones
- **Animation:** Smooth fade transitions between states (300ms)
- **API Used:**
 - **Primary Service:** `ScreenRecorderService` (Android Foreground Service)
 - **Android APIs:**
 - * `MediaProjectionManager.createScreenCaptureIntent()` - Screen capture permission
 - * `MediaProjection.getMediaProjection()` - Screen projection instance
 - * `MediaRecorder` - Video recording engine
 - * `VirtualDisplay.createVirtualDisplay()` - Screen mirroring
 - * `MediaStore` - Gallery integration for saving
 - **Permission Requirements:**

- * RECORD_AUDIO - Audio recording (optional)
- * WRITE_EXTERNAL_STORAGE - File storage
- * FOREGROUND_SERVICE - Background recording
- * POST_NOTIFICATIONS - Recording notifications (Android 13+)
- **Recording Configuration:**
 - * Format: MPEG-4 (H.264 video encoding)
 - * Frame Rate: 30 FPS
 - * Bit Rate: 8192 kbps (8 Mbps)
 - * Resolution: Device native resolution

Zoom Control Section

Zoom Selector

- **Function:** Provides 1x, 2x, and 4x zoom level selection
- **Visual States:**
 - **Enabled:** Full opacity background (AppColors.DarkBackground2), interactive zoom buttons
 - **Disabled:** 50% opacity background (DarkBackground2.copy(alpha = 0.5f)) with message
- **Disabled When:** EIS (Electronic Image Stabilization) or HDR is enabled
- **Disabled Message:** "Zoom disabled when EIS or HDR is enabled"
- **Click Action:** cameraControlViewModel.setZoom(newZoom)
- **Height:** 112dp on tablets, 56dp on phones
- **Background:** Dark background (AppColors.DarkBackground2 - #222222)
- **Zoom Button States:**
 - **Selected:**
 - * **Dark Theme:** Background: Dark gray selected (AppColors.ButtonSelectedBgColor - #515151), Text: White (AppColors.ButtonSelectedIconColor)
 - * **Light Theme:** Background: Light gray selected (#D7D7D7), Text: Dark gray (#3F3F3F)
 - **Unselected:**
 - * **Dark Theme:** Background: Medium dark (AppColors.ButtonBgColor - #333333), Text: Medium gray (AppColors.ButtonIconColor - #808080)
 - * **Light Theme:** Background: White, Text: Medium gray (#AEAEAE)
- **API Used:**
 - **Primary API:** MotocamAPIAndroidHelper.setZoomAsync()
 - **Underlying Protocol:** MotocamAPIHelper.setImgZoomCmd(zoom.displayVal)
 - **Command Structure:**

Request Packet:

Header	Command	SubCommand	Data Length	Data	CRC
SET(1)	IMAGE(4)	ZOOM(1)	1	1/2/4	1 byte

 - * Header: Header.SET (1)
 - * Command: Commands.IMAGE (4)
 - * SubCommand: ImageSubCommands.ZOOM (1)
 - * Values:
 - ZOOM.X1 (1) for 1x zoom
 - ZOOM.X2 (2) for 2x zoom
 - ZOOM.X4 (4) for 4x zoom
 - **Success Response:**

Response Packet:

Header	Command	SubCommand	Data Length	Success Flag	Data	CRC
ACK(3)	IMAGE(4)	ZOOM(1)	2	Success(0)	Success(0)	1 byte
 - **Error Response:**

Response Packet:

Header	Command	SubCommand	Data Length	Failed Flag	Error Code	CRC
ACK(3)	IMAGE(4)	ZOOM(1)	2	Failed(1)	-1 to -6	1 byte
 - **HTTP Communication:** HTTP POST to http://192.168.2.1:80/api/motocam_api

- **Zoom Levels:** Discrete levels only (no continuous zoom)

Toggle Icons Row

IMPORTANT: These are **status indicator icons only** - they are **NOT clickable/toggleable** in the Expanded Control Layout.

Stabilize Icon

- **ID:** stabilize
- **Function:** **Visual status indicator only** - shows Electronic Image Stabilization state
- **State Source:** `cameraControlState.isEisEnabled`
- **Visual States:**
 - **Active:** Selected color (white in dark theme, defined in icon config)
 - **Inactive:** Gray (`#777777`)
- **Icon Resource:** `R.drawable.git_commit_line`
- **Click Behavior:** **None** - no `.clickable` modifier applied
- **Control Location:** EIS can only be toggled via Settings → Camera Layout
- **Side Effect:** When enabled, disables zoom functionality

HDR Icon

- **ID:** hdr
- **Function:** **Visual status indicator only** - shows High Dynamic Range state
- **State Source:** `cameraControlState.isHdrEnabled`
- **Visual States:**
 - **Active:** Selected color (white in dark theme, defined in icon config)
 - **Inactive:** Gray (`#777777`)
- **Icon Resource:** `R.drawable.hd_line`
- **Click Behavior:** **None** - no `.clickable` modifier applied
- **Control Location:** HDR can only be toggled via Settings → Camera Layout
- **Side Effect:** When enabled, disables zoom functionality

View Mode Icon

- **ID:** viewmode
- **Function:** **Visual status indicator only** - shows Auto day/night mode state
- **State Source:** `cameraControlState.isAutoDayNightEnabled`
- **Visual States:**
 - **Active:** Selected color (white in dark theme, defined in icon config)
 - **Inactive:** Gray (`#777777`)
- **Icon Resource:** `R.drawable.sun`
- **Click Behavior:** **None** - no `.clickable` modifier applied
- **Control Location:** Auto day/night mode can only be toggled via Settings → Camera Layout

Implementation Details

- **Container:** Dark background box (`DarkBackground2` - `#222222`) with rounded corners
- **Layout:** Horizontal row with space between icons
- **Icon Size:** 16dp (defined in `ToggleableIcon.iconSize`)
- **No Interaction:** Icons are purely visual - no click handlers or callbacks
- **State Updates:** Icons automatically reflect state changes from `CameraControlViewModel`
- **Commented Code:** Original `onIconToggle` functionality is commented out in the codebase

Status Indicators Row

WiFi Indicator

- **Function:** Shows current WiFi connection status and signal strength
- **States:**
 - **Connected:** Active WiFi icon with signal strength indication
 - **Disconnected:** Inactive/grayed WiFi icon
- **Data Source:** `systemStatus.isWifiConnected` (passed from parent layout)
- **Signal Strength Levels:** 0-3 (Poor to Excellent)
- **API Used:**
 - **Primary APIs:**
 - * `WifiManager.getConnectionInfo()` - WiFi connection details
 - * `ConnectivityManager.getActiveNetwork()` - Network connectivity status
 - * `NetworkCapabilities.hasTransport(TRANSPORT_WIFI)` - WiFi transport verification
 - **Permission Requirements:**
 - * `ACCESS_FINE_LOCATION` or `ACCESS_COARSE_LOCATION` (Android 8.1+)
 - * Required for detailed WiFi information including SSID and signal strength
 - **Update Method:**
 - * `BroadcastReceiver` for network state changes
 - * Periodic polling every 5 seconds via `LaunchedEffect`
 - **Signal Strength Mapping:**
 - * RSSI -50 dBm → Excellent (3)
 - * RSSI -60 dBm → Good (2)
 - * RSSI -70 dBm → Fair (1)
 - * RSSI < -70 dBm → Poor (0)
 - **Fallback Logic:** Uses `ConnectivityManager` when location permission unavailable

AI Status Indicator

- **Function:** Shows AI processing status based on local configuration
- **States:**
 - **Enabled:** Blue AI icon (`StravionBlue` color)
 - **Disabled:** Gray AI icon (`Color.Gray`)
- **Data Source:** `systemStatus.isAiEnabled` (reflects local AI configuration state)
- **Configuration Management:**
 - **Local Storage:** Configuration saved to device using `Properties` file
 - **File Location:** `context.getExternalFilesDir(null)/camera_config.properties`
 - **Configuration Fields:**
 - * Object Detection (OD) - primary AI status indicator
 - * Far Object Detection (FAR)
 - * Depth Sensing (DS)
 - * Audio Processing
 - **State Updates:** Controlled via `AiConfigurationViewModel` and `CameraConfigurationManager`
- **No Camera API:** AI status is based on local app configuration, not camera hardware state
- **Padding:** Vertical 4dp
- **Icon Resource:** `R.drawable.ai_line`

Battery Indicator

- **Function:** Shows current battery level with dynamic icon and color coding
- **Display:** Battery icon with level indication (percentage hidden in this layout)
- **Data Source:** `systemStatus.batteryLevel` (0-100 range)
- **Percentage Display:** Hidden (`showPercentage = false`)
- **Color Coding:**

- **Full (80-100%):** Green (#4CAF50)
- **High (60-79%):** Light Green (#8BC34A)
- **Medium (40-59%):** Orange (#FF9800)
- **Low (20-39%):** Dark Orange (#FF5722)
- **Critical (0-19%):** Red (#F44336)
- **API Used:**
 - **Primary API:** BatteryManager Android system service
 - **Update Mechanism:** BroadcastReceiver for Intent.ACTION_BATTERY_CHANGED
 - **Battery Information:**
 - * BatteryManager.EXTRA_LEVEL - Current battery level
 - * BatteryManager.EXTRA_SCALE - Battery scale (typically 100)
 - **Registration:** Automatic receiver registration/unregistration via DisposableEffect
 - **Calculation:** (level * 100 / scale) for percentage
 - **Real-time Updates:** Immediate response to system battery broadcasts
- **Icon Selection:** Dynamic battery icon based on level ranges
- **Device Adaptation:** Icon size varies by device type (24dp tablet, 16dp phone)

Communication Architecture

HTTP Communication Layer

- **Primary Protocol:** HTTP REST API over WiFi
- **Default IP:** 192.168.2.1 (camera device)
- **Main API Port:** 80 (HTTP)
- **Health Check Port:** 8080 (ping endpoint)
- **API Endpoint:** http://192.168.2.1:80/api/motocam_api
- **Client Library:** Ktor HTTP client with CIO engine
- **Connection Management:** Automatic device discovery via ARP table
- **Wrapper Layer:** MotocamAPIHelperWrapper provides coroutine-based async operations
- **Error Handling:** Comprehensive exception handling with logging

Command Structure

All camera commands follow this structure:

[Header] [Command] [SubCommand] [DataLength] [Data] [CRC]

- **Transport:** HTTP POST with hex-encoded command data
- **Content-Type:** application/octet-stream
- **Authentication:** Session cookie authentication
- **Header:** SET(1), GET(2), ACK(3), RESPONSE(4)
- **Command:** STREAMING(1), NETWORK(2), CONFIG(3), IMAGE(4), AUDIO(5), SYSTEM(6)
- **SubCommand:** Specific operation within command category
- **Data:** Command-specific parameters
- **CRC:** Checksum for error detection

Error Codes

Error Code	Description
-1	Error in executing the command
-2	Invalid packet header
-3	Invalid command
-4	Invalid sub-command
-5	Invalid Data/Data Length
-6	CRC does not match

Health Monitoring

- **API:** `MotocamAPIAndroidHelper.getHealthStatusAsync()`
- **HTTP Endpoint:** `http://192.168.2.1:8080/ping`
- **Monitors:**
 - CPU Usage
 - Memory Usage
 - ISP Temperature
 - IR Temperature
 - Sensor Temperature
 - Service Status (RTSPS, Streamer, etc.)

Device-Specific Adaptations

Tablet Layout

- Larger padding (32dp horizontal, 22dp vertical)
- Increased spacing between elements (22dp)
- Row-based layout for main content
- Larger text sizes (20sp for header)
- Different button groupings:
 - Row 1: Settings only
 - Row 2: IR, IR Intensity
 - Row 3: Collapse Screen

Phone Layout

- Smaller padding (12dp horizontal, 14dp vertical)
- Reduced spacing (14dp)
- Column-based layout with scrolling
- Smaller text sizes (16sp for header)
- Different button groupings:
 - Row 1: IR, IR Intensity, Settings
 - Row 2: Collapse Screen

Memory Management

- Implements `DisposableEffect` for cleanup
- Calls `MemoryManager.cleanupWeakReferences()` on disposal
- Proper logging for debugging memory issues

State Management

- Uses `RecordingViewModel` for recording state
- Uses `CameraControlViewModel` for camera controls
- Uses `CameraLayoutViewModel` for layout state
- All states collected with `collectAsStateWithLifecycle()`

Animation Features

- Pulsing record dot during recording
- Smooth state transitions for record button text
- Fade animations for state changes (300ms duration)
- Scale animation for record indicator (1.0 to 1.2 scale)

This documentation covers the comprehensive Expanded Control Layout with full API integration details and visual component references.